

Lab 3 – Architecture Justification

Problem Statement #34 · Digital Gift Card & Loyalty Point Exchange | Name: _____ SRN: _____

Architecture Selection

We chose **Microservices Architecture** for the **Digital Gift Card & Loyalty Point Exchange System**.

Architectural Choice – Style Comparison

Style	Pros for this scenario	Cons for this scenario	Verdict
Layered	Simple, well understood	Whole app scales as one unit; a merchant-sync spike can slow redemption; one shared DB is a single trust zone	Rejected
Client-Server	Central control, easy consistency	Single point of failure; bottleneck at peak load; weak fault isolation	Rejected
Microservices	Independent scaling, fault and security isolation, replaceable adapters	Network latency, operational complexity, consistency risk (contained – see below)	Selected

Two Scenario-Specific Reasons

- **1. Very different workloads need independent scaling and fault isolation.** Latency-critical conversion and redemption (250 ms, NFR-001), dashboard reads (2 s, FR-002) and merchant point sync (24 h tolerance, FR-005) behave very differently. As separate services, Conversion, Voucher and Wallet scale on their own, and an outage or slow response from a Merchant Partner only affects the Merchant Integration Service, not redemption.
- **2. Volatile external integrations must be replaceable.** Merchant Partners and the Payment Gateway use different protocols and change independently. Wrapping each in an adapter component (Merchant Integration Service, Payment Gateway Adapter) behind a stable interface (IMerchantSync, IFulfilment) lets us swap a provider without touching the core conversion and voucher logic, matching the “replaceable, black-box component” principle.

Security Advantage

Security-critical data is isolated by service. Only the Ledger & Audit Service can reach the Ledger DB (AES-256 at rest), only the Voucher Service handles voucher codes, and every request enters through one API Gateway (OAuth2/JWT, TLS 1.3, rate limiting). Internal calls use mTLS, so a compromised adapter cannot read vouchers or alter balances directly. Every conversion and redemption is appended to a hash-chained, tamper-evident audit log (NFR-002), and single-use vouchers are locked at the Voucher Service, preventing double-spend (FR-004).

Performance Benefit

The latency-critical path (Gateway → Conversion → Wallet → Ledger → Voucher) uses lightweight in-cluster gRPC calls and only the services it needs. Slow merchant syncing runs asynchronously through a message queue, off the hot path, and the Conversion and Voucher services can be scaled out at peak load without scaling the whole platform. This helps meet the 250 ms target in NFR-001.

Trade-off Acknowledged

Microservices risk data inconsistency. To keep “zero ledger balance drift”, all balance changes go through a single-writer Ledger Service with ACID transactions and idempotency keys, so no distributed transaction is needed for balances. Multi-step flows (convert → issue voucher) are handled as compensating steps.